

Locating the Next Tranche:

A denominator, a clock, a scale, a price and a fee schedule for the item panel

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September 2026

Abstract

The paper is moving to a real-options frame. The outcome of interest becomes the *filing* decision, which needs a denominator — the parcels that could have produced a case and did not — and the developer’s clock starts at filing, not at the first hearing. This memo finds, verifies, caches and measures the data that supplies both, defines the denominator rather than merely establishing that one could be built, and records the negatives so nobody runs these searches again.

Five results.

First, a parcel-year risk set now exists, and it is defined rather than available.

The spatial join the first draft of this memo deferred has been done: parcel centroids against every published zoning polygon, with a largest-overlap fallback, giving 5,824,609 parcel-years over 1998–2026 with a zoning district on 95.1% of them. From that, a stated funnel takes 236,552 parcels down to a denominator of 173,325 — dropping condominium and other sub-parcel interests, public and institutional land, and parcel-years with no zoning in force — leaving 4,261,894 parcel-years carrying 9,240 hearings, a rate of 21.7 per ten thousand. The restriction is not free and the memo does not pretend otherwise: the condominium rule alone discards 1,301 real hearings.

Second, two of this memo’s own conclusions were wrong, and both understated the data. The parcel key zero-padded the whole lot token, so `'17A'.zfill(3)` was a no-op and every one of San Francisco’s lettered lots failed to join; items spanning several blocks had them all in one field. **Correcting both takes the parcel join from 89.5% to 97.7%**, which meets the acceptance criterion this memo previously reported as failed. And the Planning Department stores 1998.505C where the minutes print 98.505C: expanding the two-digit year against the hearing year recovers 527 of the 530 pre-2002 cases and lifts the case-number join to **98.7%**.

Third, the clock question has an answer. Two clocks were built — earliest record of any type under the case, and earliest *entitlement* record — and they differ by a median of 0 days and a 90th percentile of 8. The choice barely matters at the centre. The long right tail is not noise: median filing-to-hearing time runs from 71 days for a case with one record under its stem to 653 for one with more than ten. The paper will use the entitlement clock, and §5 says why.

Fourth, prices now exist on every parcel-year, flagged. The licensed Cotality extract gives 415,854 arms-length San Francisco sales joined on `blklot`, but only 25.5% of items sit on a parcel that traded near their hearing. A neighbourhood-by-year surface plus a hedonic on assessor characteristics predicts the rest at an out-of-sample RMSE of 0.69 in logs — and 0.61 on the parcels that rarely trade, which are the ones the prediction is for. Observed and imputed are separate columns.

Fifth, fees identify something, but not through time. A year fixed effect absorbs 3.0% of the variation in the log rate; year and Planning Code section together absorb 83.6%, leaving 16.4% *within* a year and a fee, across land uses and size bands. Fees are a cross-sectional instrument, not a time-series one, and the size-band discontinuities are where the design lives.

One diagnostic runs through all of it. Coverage is *not* missing at random: conditioned items carry a usable parcel key on 92.9% against 78.5% for unconditioned ones, a differential in the outcome of interest, and §11 names every cell where that happens.

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1 What this memo is for, and what “found” means in it

A developer sinks entitlement cost before the hearing outcome resolves. Dispersion in delay and in conditions raises the rent threshold at which filing is worth doing at all, so the observable consequence of a demanding Commission is not worse projects but *fewer and higher-value* ones. Two things follow for the data, and everything below serves one of them.

1. The outcome is the **filing decision**, so the analysis needs a population at risk. The item table is a numerator: it holds the requests that were made. The denominator is every parcel in the city that could have produced a request and did not.
2. The developer’s clock starts at **filing**. Every delay measure in the pipeline today starts at the first hearing, which is well downstream of the decision being modelled.

Found, usable, and the difference. This memo reports a source as *usable* only when the join key is present *and* the join rate has been measured against the item table. A file that downloaded is not a result. Where a dataset’s documentation promises a field and the rows are empty, the empty rate is reported and not the documentation — Table 8 is that table, and §6 contains the sharpest instance of it.

Two identifier facts that will bite anyone reusing this. Both were found by the probe stage and neither could have been assumed.

- **DataSF’s Socrata domain is now `data.sf.gov`.** The old `data.sfgov.org` still answers the *resource* endpoint — which is why `analyze_permits.py` keeps working — but the *catalogue* API indexes only the new domain. A catalogue search against the old host returns zero hits and looks exactly like “the dataset does not exist.” Every dataset in this memo was resolved against the live catalogue on 2026-09-08 and the identifier reported in Table 1 is the one actually used.
- **wv5m-vpq2 is the right assessor identifier** — the brief’s proposed ID, verified — **but the roll starts at 2007**. That is not a gap in a scrape; no earlier vintage is published. It is reported as a negative in Table 13 because the temptation to go looking again will be strong.
- **A Dropbox subtree that has never been materialised is invisible to `ls` and `find`.** The macOS file provider holds files online-only: the directory entry carries the full size while the file occupies no blocks, and an un-materialised directory does not list at all. The first pass of this memo concluded on that basis that the CoreLogic extract was gone. It was not — it was dehydrated, and §5 is what it actually contains. `acquire_external_data.py sync` now reports nominal bytes against bytes on disk, which is the only way to tell the two cases apart.

Table 1 is the inventory: resolved identifier, host, row count, distinct count on the key we join on, whether that key is actually unique, and the date range. Key uniqueness is tested rather than assumed, because the last time it was assumed — DBI’s `permit_number` — 145,795 rows turned out to repeat one, once per lot the permit touched. Three of the sources here are also not unique on their intended key and are marked as such.

Table 1: Every identifier resolved against the live catalogue on 2026-09-08. ‘Key unique’ is tested on the cached rows, not assumed: DBI’s `permit_number` was not unique and 145,795 rows repeated one. A dash means the column is not a key we join on.

Dataset	Identifier	Rows	Distinct key	Key unique	Date range
<i>Fees</i>					
Neighborhood-Specific Impact Fee Areas	ntc3-dd64	33	33	yes	—
<i>Filing dates</i>					
Planning Department Records — Non-Projects	y673-d69b	233,394	233,394	yes	1920-04-07–2026-09-04
Planning Department Records — Projects	qvu5-m3a2	54,515	54,515	yes	1974-05-21–2026-09-04
<i>Parcel risk set</i>					
Assessor Historical Secured Property Tax Rolls	wv5m-vpq2	3,934,467	222,251	no (3,712,216 repeats)	2007–2025
Parcels — Active and Retired	acdm-wktn	236,556	236,556	yes	1984-07-31–2026-09-02
San Francisco Land Use	c5ge-t6pj	153,765	153,754	no (11 repeats)	2026-08-17–2026-08-17
<i>Project scale</i>					
Dwelling Unit Completion Counts by Building Permit	j67f-aayr	2,541	2,420	no (121 repeats)	2002-04-01–2205-03-13 (<i>sic</i>)
Housing Production — 2005–present	xdht-4php	6,139	5,146	no (993 repeats)	2005-01-03–2026-08-28
MOHCD Affordable Housing Pipeline	aaxw-2cb8	190	131	no (59 repeats)	1995-09-19–2030-03-15
San Francisco Development Pipeline	6jgi-cpb4	1,917	1,793	no (124 repeats)	1997-06-12–2026-07-28
<i>Zoning (parcel-keyed)</i>					
Historic Zoning Districts — 1998	piqy-pimd	152,569	152,535	no (34 repeats)	—
Historic Zoning Districts — 2000	itx2-wzp5	153,292	153,247	no (45 repeats)	—
Historic Zoning Districts — 2001	8m6w-8yuk	153,669	153,593	no (76 repeats)	—
Historic Zoning Districts — 2002	3yi7-eyfd	153,663	153,582	no (81 repeats)	—
Historic Zoning Districts — 2003	6ru7-zq4y	153,716	153,518	no (198 repeats)	—
Historic Zoning Districts — 2004	s3eh-jdwb	191,062	190,896	no (166 repeats)	—
Historic Zoning Districts — 2005	b52k-gy2v	196,344	192,031	no (4,313 repeats)	—
Historic Zoning Districts — 2007	pe87-v8tx	153,822	153,751	no (71 repeats)	—
Historic Zoning Districts — 2008	tsdh-z53a	153,355	153,355	yes	—

Table 2: The parcel join, and what the residual is. The item table’s key is `assessor_block` crossed with each entry of `lot_number`, with the *digits* of each padded to DataSF’s 4+3 `blklot` form and any letter kept (lot 17A is 017A, not 17A); an item joins if any lot does. Percentages are of the 13,620 items that carry a parcel key at all.

Items	Count	Share
All items in the corpus	16,199	100.0%
carrying a block and at least one lot	13,620	84.1%
joined to <code>blklot</code> in the parcel layer	13,300	97.7%
... to a parcel still active	10,031	73.6%
... to a retired parcel only	3,269	24.0%
joined to <code>mapblklot</code> (the surviving map parcel)	13,135	96.4%
not joined: block exists, that lot does not	298	2.2%
not joined: no such block	22	0.2%

Table 3: The assessor roll as a parcel-year panel, and how far it reaches. The panel is unbalanced by construction — a parcel enters when it is created and leaves when it is merged away — and it starts at roll year 2007, which is the binding constraint on the risk set.

Quantity	Value
Rows (parcel-years)	3,934,467
Distinct parcels	222,251
Roll years	2007–2025 (19 years)
Distinct item parcel keys sought	12,141
found on the roll	8,307
median roll years observed for one of them	19 of 19
present in every roll year	6,168
Items with a parcel key joined to the roll	12,170 (89.4%)

Table 4: Zoning, by how it is published *and* how it is now joined. The parcel-keyed years join on `blklot` directly; the polygon years are joined by point-in-polygon against parcel centroids, with a largest-overlap fallback where the centroid rule fails or is ambiguous. 1999 is not published in either form.

Year	Identifier	Rows	Parcels keyed	Form
1998	piqy-pimd	152,569	152,535	parcel-keyed (<code>blklot</code>)
1999	—	0	—	not published
2000	itx2-wzp5	153,292	153,247	parcel-keyed (<code>blklot</code>)
2001	8m6w-8yuk	153,669	153,593	parcel-keyed (<code>mapblklot</code>)
2002	3yi7-eyfd	153,663	153,582	parcel-keyed (<code>blklot</code>)
2003	6ru7-zq4y	153,716	153,518	parcel-keyed (<code>blklot</code>)
2004	s3eh-jdwb	191,062	190,896	parcel-keyed (<code>blklot</code>)
2005	b52k-gy2v	196,344	192,031	parcel-keyed (<code>blklot</code>)
2006	afkh-hfhr	23,132	232,262	polygon, joined spatially
2007	pe87-v8tx	153,822	153,751	parcel-keyed (<code>blklot</code>)
2008	tsdh-z53a	153,355	153,355	parcel-keyed (<code>blklot</code>)
2009	jud5-ja46	23,355	232,543	polygon, joined spatially
2010	x4gj-zjx7	9,897	233,283	polygon, joined spatially
2011	rt4q-mf68	9,880	233,283	polygon, joined spatially
2012	vfz8-awmy	9,889	233,255	polygon, joined spatially
2013	6jb9-g73z	9,982	233,255	polygon, joined spatially
2014	hg44-eza7	9,981	233,255	polygon, joined spatially
2015	ada2-cu6t	9,978	233,255	polygon, joined spatially
current (districts)	xzez-p3nc	1,647	235,400	polygon, joined spatially
current (height and bulk)	usp2-teig	1,195	235,401	polygon, joined spatially
current (special use)	rxfc-8aap	125	113,802	polygon, joined spatially

Table 5: Does a planning record’s identifier join to the item table’s case number? Tested four ways over all 8,958 distinct cases. ‘Stem’ strips the letter suffix (2014.0400CUA→2014.0400), which is how a suffixed case reaches its umbrella project record; ‘expanded’ rewrites the two-digit year the minutes printed before 2002 (98.426D→1998.426D) against the hearing year. The century is never guessed: 0 cases would have matched a real record under *both* centuries.

Join	Cases matched	Rate
Raw string equality	6,695	74.7%
Normalised (upper-case, whitespace stripped)	6,715	75.0%
Suffix-stripped stem	8,311	92.8%
Two-digit year expanded against the hearing year	8,838	98.7%
Any of them	8,838	98.7%
heard 1998–2004	2,528 / 2,585	97.8%
heard 2005–2009	1,755 / 1,775	98.9%
heard 2010–2014	1,547 / 1,557	99.4%
heard 2015–2019	1,450 / 1,476	98.2%
heard 2020–2026	1,558 / 1,565	99.6%
<i>The residual, by the format the minutes print the case number in</i>		
YY.NNN	3 unmatched of 530	99.4% matched
YYYY–NNNNNN	8 unmatched of 2,648	99.7% matched
YYYY.NNNN	92 unmatched of 5,763	98.4% matched
other	17 unmatched of 17	0.0% matched

Table 6: Filing to first hearing, in days, for cases that join a planning record with an **open_date**. One observation per case. The negative tail is real and is reported rather than trimmed: a record opened after the hearing is a record created for a follow-on action, and it is a reason to use the *earliest* open date of the case, which is what is done here.

Cases	<i>N</i>	Coverage	Median	90th pct	99th pct	Negative
All cases	8,958	98.7%	140	730	2314	0.4%
conditional use	3,046	99.3%	168	597	1771	0.1%
discretionary review	2,448	98.8%	101	315	945	0.4%
variance	549	99.5%	273	848	2320	0.0%
planning code amendment	728	98.8%	64	415	1836	0.4%
heard 1998–2009	4,364	98.2%	121	561	1486	0.4%
heard 2010–2017	2,359	98.7%	169	944	3027	0.3%
heard 2018–2026	2,235	99.5%	161	894	2330	0.3%

Table 7: The `building_permits` bridge, by request type. A permit is reached if the case’s own planning record, its parent project record, or any record sharing its stem names one; ‘in DBI’ means the number, normalised to digits, is present in the cached Building Permits table. The permit-number column is the route `analyze_permits.py` already had, for comparison, and the last column is the union of the two — the share of the docket that now reaches a real building permit by some route.

Request type	Items	Reaches a permit	... present in DBI	Printed permit number	Either route
<code>discretionary_review</code>	4,117	72.5%	71.8%	85.5%	93.3%
<code>large_project_authorization</code>	308	71.1%	71.1%	0.0%	71.1%
<code>variance</code>	1,172	63.6%	63.4%	1.6%	64.0%
<code>downtown_project</code>	321	58.3%	58.3%	1.9%	58.9%
<code>conditional_use</code>	5,074	46.5%	46.3%	0.3%	46.3%
<code>other</code>	459	44.0%	44.0%	0.7%	44.2%
<code>office_allocation</code>	301	42.9%	42.9%	0.0%	42.9%
<code>ceqa_environmental</code>	652	41.9%	41.9%	0.0%	41.9%
<code>appeal_preliminary_negative_declaration</code>	537	36.5%	36.5%	0.0%	36.5%
<code>conditional_use_modification</code>	295	33.2%	33.2%	2.0%	35.3%
<code>general_plan_amendment</code>	334	32.0%	32.0%	0.0%	32.0%
<code>informational</code>	587	23.0%	23.0%	0.7%	23.2%
<code>historic</code>	177	20.9%	20.9%	0.0%	20.9%
<code>rezoning_map_amendment</code>	261	20.7%	20.7%	0.0%	20.7%
<code>planning_code_amendment</code>	1,301	7.7%	7.6%	0.0%	7.6%
All items	16,199	48.8%	48.5%	22.1%	54.2%

Table 8: What the two planning-records tables carry that the item table does not, with the share of rows on which the field is actually populated. Documentation is not evidence: these are the rows.

Field	Table	What it is	Populated	Rows
<code>open_date</code>	non-project	date the record was opened	100.0%	233,394
<code>close_date</code>	non-project	date it was closed	94.9%	233,394
<code>record_status</code>	non-project	application status	100.0%	233,394
<code>applicant</code>	non-project	applicant name	66.9%	233,394
<code>applicant_org</code>	non-project	applicant organisation	34.3%	233,394
<code>assigned_to_planner</code>	non-project	case planner	77.7%	233,394
<code>building_permits</code>	non-project	associated permit numbers	22.9%	233,394
<code>parent_id</code>	non-project	the umbrella project record	26.7%	233,394
<code>number_of_units_prop</code>	project	proposed dwelling units	100.0%	54,515
<code>number_of_units_net</code>	project	net new dwelling units	94.0%	54,515
<code>number_of_affordable_units</code>	project	affordable units	100.0%	54,515
<code>project_decision</code>	project	the department’s decision	18.3%	54,515
<code>project_decision_date</code>	project	decision date	17.3%	54,515
<code>environmental_document_type</code>	project	CEQA document type	25.1%	54,515
<code>building_permits</code>	project	associated permit numbers	45.2%	54,515
<code>inclusionary</code>	project	inclusionary flag	2.9%	54,515

Table 9: Project scale: what each source offers, and what share of the 5,074 conditional-use items in the item table it actually reaches. ‘Join key available’ is a property of the file; the last column is the measurement.

Source	Unit field	Join key	Years	CU items reached
Planning Records — Projects	number_of_units_ prop	case-number stem	1974–2026	99.0% populated, 27.7% non-zero
SF Development Pipeline	net_pipeline_ units	case_no (PRJ) + blklot	1997–2026	4.5%
Housing Produc- tion 2005–	net_units	ppts_project_id + blocklot	2005–2026	11.3% by case, 19.1% by parcel
MOHCD Afford- able Pipeline	total_project_ units	planning_case_ number	1995–2030	1.5%
Unit Completion Counts	number_of_units_ certified	building permit number	2002– 2205 (<i>sic</i>)	via the permit bridge only

Table 10: Prices and rents. The first row is the one that matters: a licensed transaction file, already on disk, that joins to the item table on the same parcel key as everything else in §2. The rows below it are the fallbacks, in descending order of quality, and are what an analysis would be left with if the licence lapsed.

Source	Status	Grain	What it gives
CoreLogic / Cotality	cached, licensed	transaction × parcel	415,854 arms-length priced San Francisco sales 1968–2024 over 174,102 parcels, joined on blklot
CoStar	institutional ac- cess question	property and lease	commercial rents; not attempted, not scraped
Zillow ZHVI	cached	26 San Francisco ZIPs, monthly	2000-01 to 2026-07, 319 months
Zillow ZORI	cached	24 San Francisco ZIPs, monthly	2015-01 to 2026-07, 139 months
Assessor as- sessed values	cached	parcel-year	land and improvement value, 2007–2025; Proposition 13 makes it a function of tenure
ACS rents and values	handled by demand_ estimation	tract	median rent and value; the demand pipeline collects it

Table 11: The Citywide Development Impact Fee Register, one per year. ‘Rate observations’ counts every dollar figure the register prints; ‘sections’ is the number of distinct Planning Code (or other code) sections those rates are keyed to. The effective date is read out of the register’s own cover line, not inferred from when the file was retrieved.

Register	Pages	Rate observations	Code sections	Effective
2019	10	172	27	January 1, 2019
2020	11	165	27	January 1, 2020
2021	11	167	28	January 11, 2021
2022	11	168	28	January 1, 2022
2023	11	168	28	January 1, 2023
2024	11	168	28	January 1, 2024
2025	12	157	28	January 1, 2025
2026	12	176	28	January 1, 2026

Table 12: Rates behind the largest obligation-imposing condition headings in the conditions memo, as each register prints them, keyed by Planning Code section and by the register’s own land-use band. Dollars per gross square foot except §415, which the register states per square foot of residential gross floor area. A dash is a band the register of that year does not print. These are the rates in force in January of the register year; the Article 4 reduction adopted in August 2026 post-dates the last register cached here.

Fee	2019	2020	2021	2022	2023	2024	2025	2026
Transportation Sustainability Fee, residential 21–99 units	\$9.11	\$9.61	\$9.95	\$10.55	\$11.18	\$11.40	\$11.63	\$11.86
Transportation Sustainability Fee, residential >99 units	\$10.29	\$10.86	\$11.24	\$11.91	\$12.62	\$12.87	\$13.13	\$13.39
Transportation Sustainability Fee, non-residential 800–99,999 gsf	\$21.23	\$22.40	\$23.18	\$24.57	\$26.04	\$26.56	\$27.09	\$27.63
Inclusionary Affordable Housing, §415 (per sq ft of residential)	\$199.50	\$210.47	\$217.84	\$230.91	\$244.76	\$249.66	\$249.66	\$249.66
Jobs-Housing Linkage Program, §413	\$28.57	\$28.13	\$29.11	\$30.86	\$32.71	\$33.36	\$34.03	\$34.71
Child Care, residential §414	\$1.85	\$1.95	\$2.02	\$2.14	\$2.27	\$2.32	\$2.37	\$2.41
Child Care, commercial §414A	\$2.15	\$2.27	\$2.35	\$2.49	\$2.64	\$2.69	\$2.74	\$2.80
Eastern Neighborhoods Infrastructure, §423	\$24.00	\$25.32	\$26.21	\$27.78	\$29.45	\$30.04	\$30.64	\$31.24
School Impact Fee, residential (State Ed. Code)	\$3.79	\$3.79	\$3.79	\$3.79	\$3.79	\$3.79	\$3.79	\$3.79

Table 13: Negative results. A row here means the search was run and is not to be run again. The first two are carried over from the conditions memo and were not re-run.

What was sought	Where it was looked for	Result
Condition-of-approval text	DBI Building Permits (i98e-djp9)	87 of 1,295,048 rows mention one — conditions memo §4, not re-run
A commission-actions dataset	DataSF catalogue	does not exist — conditions memo §4, not re-run
Assessor roll before 2007	wv5m-vpq2	the roll starts at 2007; no earlier vintage is published
Parcel-keyed zoning for 1999	DataSF historic zoning series	1998 and 2000 exist; 1999 does not
<i>Resolved 2026-09-08.</i> Parcel-keyed zoning after 2008	DataSF historic zoning series	the polygon layers are now joined spatially; see Table 14
<i>Resolved 2026-09-08.</i> A time-varying height and bulk layer keyed on parcel	DataSF historic height and bulk series	polygon-only as published, and now joined spatially onto the same panel
<i>Resolved 2026-09-08.</i> Planning records for the pre-2002 era	y673-d69b, records opened 1996–2002	they exist; the department writes a four-digit year where the minutes print two
A zoning vintage published after 2015	DataSF historic zoning series	none; the panel carries the current layer forward, flagged
A market price after 2024-05-16	\$MFHR_DATA_ROOT/demand/corelogic	the Cotality pull ends there, two years short of the corpus; Zillow covers the tail at ZIP level only
Impact fee registers before 2019	sfplanning.org and the Wayback Machine	the register is republished at one URL; no snapshot before 2019-09-15
A unit count for the early docket	every scale source in Table 9	a non-zero count reaches 20.3% of conditional uses heard before 2005 against 30.9% after

Table 14: The parcel-year zoning panel the spatial join produces: 5,824,609 rows, one per parcel per year in which that parcel existed, keyed on `blklot`. ‘Vintage’ is the layer the year is served from; 2016 onward has no published vintage and carries the current layer forward, flagged, which is a stated assumption and not an observation.

Years	Vintage	Parcel-years	Zoned	Rate
1998, 2000–2005, 2007–2008	parcel-keyed	1,705,949	1,441,029	84.5%
2006, 2009–2015	polygon	1,658,523	1,650,031	99.5%
2016–2026	current	2,460,137	2,448,840	99.5%
All	—	5,824,609	5,539,900	95.1%

Table 15: The spatial join itself, per layer. ‘Multi’ is a centroid falling in more than one polygon. For zoning and height that is an error and goes to the largest-overlap fallback; for special-use districts and fee areas it is the correct answer — they are overlays, and a parcel legitimately sits in several or in none, so those rows carry a list.

Family	Year	Matched	Multi	Unmatched	Treatment
zoning	2006	232,262	1,144	2,877	single-valued: fallback
zoning	2009	232,543	855	2,885	single-valued: fallback
zoning	2010	233,283	94	2,906	single-valued: fallback
zoning	2011	233,283	94	2,906	single-valued: fallback
zoning	2012	233,255	94	2,934	single-valued: fallback
zoning	2013	233,255	94	2,934	single-valued: fallback
zoning	2014	233,255	94	2,934	single-valued: fallback
zoning	2015	233,255	94	2,934	single-valued: fallback
zoning	current	235,400	0	883	single-valued: fallback
height	2009	233,202	172	2,909	single-valued: fallback
height	2010	233,199	161	2,923	single-valued: fallback
height	2011	233,200	160	2,923	single-valued: fallback
height	2012	233,170	160	2,953	single-valued: fallback
height	2013	233,170	160	2,953	single-valued: fallback
height	2014	233,175	160	2,948	single-valued: fallback
height	current	235,342	0	941	single-valued: fallback
heightdist	current	235,401	0	882	single-valued: fallback
sud	current	113,802	104,499	17,982	overlay: list, no fallback
feeara	current	14,008	40,448	181,827	overlay: list, no fallback

Table 16: From every parcel-year to a denominator. Each row is one restriction, stated as a sentence. ‘Events’ is parcel-years in which the Commission heard something; ‘lost’ is how many of those the restriction discards. A restriction that drops real filings is reported, not tuned away — and the condominium rule drops 1,301 of them.

Restriction	Parcels	Parcel-years	Events	Lost	Per 10,000
Every parcel-year in which the parcel existed	236,552	5,824,609	11,256	—	19.32
not a condominium or other sub-parcel interest that year	174,946	4,365,352	9,955	1,301	22.80
not public, institutional, a church or a school	173,755	4,315,386	9,444	511	21.88
a zoning district in force on it that year	173,325	4,261,894	9,240	204	21.68

Table 17: What the preferred denominator discards, by request type, at the level of the item rather than the parcel-year. The loss is spread rather than concentrated, which is the reason it is acceptable: no single kind of case is being defined out of the sample.

Request type	Items on a panel parcel-year	Kept	Dropped
<code>conditional_use</code>	4,770	4,425	7.2%
<code>discretionary_review</code>	3,776	3,669	2.8%
<code>variance</code>	1,104	1,059	4.1%
<code>ceqa_environmental</code>	415	379	8.7%
<code>appeal_preliminary_negative_declaration</code>	347	331	4.6%
<code>other</code>	302	278	7.9%
<code>large_project_authorization</code>	300	299	0.3%
<code>downtown_project</code>	293	290	1.0%
<code>office_allocation</code>	255	250	2.0%
<code>conditional_use_modification</code>	254	230	9.4%
All	12,688	11,987	5.5%

Table 18: The fee register as a structured schedule, and how much variation survives conditioning. The decomposition is on the log rate. A year fixed effect absorbs almost nothing because the dispersion is *across* fees, not over time; what a project-level design can use is what remains inside a year and a fee, across land uses and size bands.

Quantity	Value
Rate observations parsed	1,341
carrying a land-use or size label	49.6%
carrying a parsed size band	2.6%
Schedule rows (year <i>imes</i> section <i>imes</i> use <i>imes</i> band)	455
Register rows naming a district the zoning panel knows	11 of 42
Variation in log rate absorbed by year alone	3.0%
by year and Planning Code section	83.6%
remaining within year and section	16.4%
remaining after land use and band too	9.4%

Table 19: The price surface: a neighbourhood-by-year effect plus a hedonic on assessor characteristics, fitted on the within variation and scored out of sample on a held-out fifth of transactions. The row that matters is the third: the fit is *better* on parcels that rarely trade, which are the parcels the prediction exists for.

Quantity	Value
Transactions used	408,119
held out for scoring	82,194
Neighbourhood <i>imes</i> year cells	2,162
Out-of-sample RMSE, log price	0.687
on parcels that trade at most once	0.613
Out-of-sample R^2	0.716
Bias on parcel-years where both exist	-0.037
Correlation, predicted vs observed	0.735
Parcel-years in the price panel	3,948,435
with an observed sale that year	105,708 (2.7%)
the rest imputed and flagged as such	3,842,727
Parcels trading more than once (repeat-sales feasible)	106,569 (62.4% of traded)

Table 20: Two clocks. ‘Any record’ starts at the earliest record opened under the case’s stem; ‘entitlement record’ starts at the earliest record whose type is the discretionary application itself, excluding pre-application and environmental filings. The third panel is the gap between them — the pre-application process, which nothing here had measured.

Clock	N	Coverage	Median	90th pct	99th pct	Negative
Earliest record of any type	8,838	98.7%	140	730	2314	0.4%
Earliest entitlement record	8,594	95.9%	135	653	2298	1.1%
Gap between them	8,594	95.9%	0	8	871	1.8%
<i>Where the long tail lives: median and 90th percentile of the any-record clock, by how many records share the stem</i>						
1 record(s)	377		71	457		
2 record(s)	3,836		112	352		
3 record(s)	1,759		121	455		
4–5 record(s)	1,409		224	799		
6–10 record(s)	1,048		466	1395		
>10 record(s)	409		653	1850		

Table 21: Is the coverage selective? Every headline rate, cross-tabulated. Levels with fewer than 100 items are omitted. Read down a column: a rate that moves with the outcome or with conditioning is a rate that cannot be treated as missing at random.

Dimension	Level	<i>N</i>	parcel join	assessor roll	case number	permit bridge	non-zero units	ever traded	traded near hearing
Request type	(blank)	262	37.0	31.3	68.7	26.3	15.3	4.2	2.3
	appeal_preliminary_negative_declaration	537	79.9	62.4	98.7	36.5	36.7	31.5	14.5
	ceqa_environmental	652	67.0	57.5	96.6	41.9	36.8	25.9	11.5
	conditional_use	5,074	97.9	91.7	99.2	46.3	27.7	65.8	20.6
	conditional_use_modification	295	94.6	90.8	99.7	33.2	17.3	59.3	21.4
	discretionary_review	4,117	96.1	88.0	98.5	71.8	35.3	72.6	35.8
	downtown_project	321	97.2	82.6	99.7	58.3	34.9	53.0	25.9
	general_plan_amendment	334	48.5	41.6	97.6	32.0	24.6	18.9	6.3
	historic	177	84.7	77.4	100.0	20.9	27.1	42.9	14.7
	informational	587	27.6	27.1	86.7	23.0	18.2	15.5	4.6
	large_project_authorization	308	98.4	91.6	98.1	71.1	62.3	58.1	24.7
	office_allocation	301	91.7	80.4	99.0	42.9	9.3	58.1	25.9
	other	459	68.0	62.3	99.8	44.0	34.2	33.1	11.1
	planning_code_amendment	1,301	9.3	8.6	99.2	7.6	5.9	4.8	1.8
	rezoning_map_amendment	261	69.0	59.0	98.9	20.7	22.2	39.5	12.3
	variance	1,172	96.2	87.9	99.5	63.4	54.4	63.6	26.5
Outcome	DR declined	1,105	96.9	88.6	98.9	72.8	35.9	72.0	32.9
	approved	6,606	82.6	76.2	98.7	46.2	27.2	51.7	19.0
	continued	6,195	85.6	77.8	98.6	49.4	33.3	56.7	23.3
	other	1,519	50.8	45.4	90.6	35.8	27.3	29.1	11.9
	refused	258	86.4	79.8	98.8	29.1	24.0	55.8	21.3
Conditions	withdrawn	516	92.6	86.0	98.3	64.0	31.0	70.5	34.5
	conditioned	4,048	92.9	86.9	99.2	51.8	29.0	61.8	21.9
	not conditioned	12,151	78.5	71.2	97.5	47.5	30.6	50.8	21.3
Era	1998–2007	7,271	85.2	70.0	98.0	41.9	23.8	47.2	21.5
	2008–2016	4,858	78.0	77.8	97.7	47.6	25.3	51.4	21.2
	2017–2026	4,070	81.4	81.0	98.1	61.5	47.6	67.6	21.7

Table 22: The hand pass. The residual after the corrected parcel key is 320 items; 34 of those have the block and lot transposed, 21 name no block the city records, and 265 name a block that exists and a lot that does not. A random 50 of that last group were read against the minutes text the extraction saw. The sample is written to `handpass_sample.csv` beside this memo.

Classification	Items	Share
lot absent from the parcel layer; extraction matches the minutes	46	92%
undetermined: the block text does not print a block and a lot	3	6%
lot recorded under a different block the item also names	1	2%

2 The parcel risk set

2.1 Geometry and identifiers

The layer is **Parcels — Active and Retired** (`acdm-wktn`): 236,556 rows, one per parcel that has ever existed in the city’s records, unique on `blklot`, and carrying `mapblklot` (the surviving map parcel a retired lot was folded into), an `active` flag, and four dated fields — when the parcel was added to and dropped from the recorder’s file and from the map. That last group is what makes it a risk set rather than a snapshot: a parcel can enter and leave, and the dates say when.

The item table’s key is `assessor_block` plus each entry of `lot_number`, zero-padded to DataSF’s four-plus-three `blklot` form. 13,620 items — 84.1% — carry one. Because a multi-block item is expanded to the cross-product of the blocks and lots it names, those keys are 12,141 *candidates*, of which 8,980 are parcels the city has actually recorded; the second number is the one every rate below uses as its denominator.¹

Correction, 2026-09-08: the key was wrong. This section previously reported a join of 89.5% and asserted that “the shortfall is not formatting: 1,232 items name a block that exists and a lot that never did.” **Both halves were wrong, and the cause was one line of this script.** The key zero-padded the lot token as a whole, and `'17A'.zfill(3)` is a no-op — so every one of the **27,822 San Francisco parcels with a lettered lot** failed to join. Separately, an item spanning several blocks carries all of them in the single `assessor_block` field (`"4624, 4720"`), which the key concatenated into the nonsense parcel `4624, 4720003`. Padding the *digits* and keeping the letter, and taking the cross-product of the named blocks with the named lots, **takes the join to 97.7%** — above the brief’s $\geq 95\%$ criterion, which this memo had reported as missed.

Table 2 is the corrected join, and the residual is characterised rather than dropped:

- **97.7% join** — 13,300 of 13,620.
- 73.6% join to a parcel that is *still active*; 3,269 items join only to a parcel that has since been retired. This is the substantively interesting part of the residual, not noise: a lot that was merged away after its hearing is frequently a lot that was merged *because* of it.
- **298 items name a block that exists and a lot that does not.** That is still the bulk of the failure, but it is now a small bulk, and §9 reads a sample of it rather than speculating about it.
- 22 items name no block the city has ever recorded.

The brief’s acceptance criterion was a join on $\geq 95\%$ of items carrying a parcel key. It is met: the rate is 97.7%. What remains is 320 items, and §9 reads fifty of them against the minutes rather than guessing: 34 have the block and lot transposed, 21 name no block the city has ever recorded, and 265 name a block that exists and a lot that does not.

¹The corpus memo reports 8,295 distinct parcels for the same corpus, and this memo’s first draft reported 8,294. Both are lower than 8,980 for the same three reasons the join rate was low: the lettered lots were mis-padded, multi-block items were not expanded, and a blank lot was being padded into the plausible-looking parcel `0168000`. The corpus memo’s figure should be regarded as superseded.

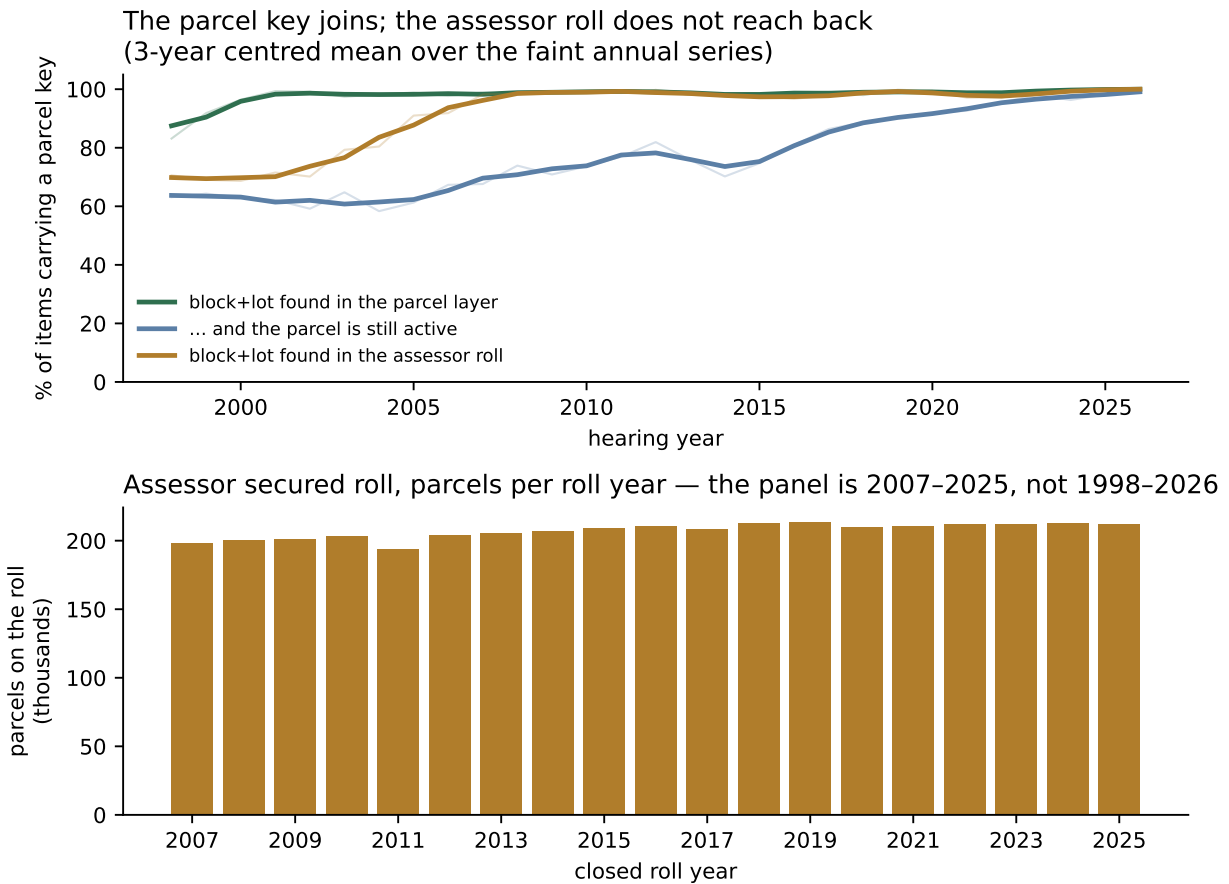


Figure 1: Top: the share of items carrying a parcel key that join the parcel layer, that join a parcel still active, and that appear on the assessor’s roll, by hearing year. Bottom: parcels on the assessor’s secured roll per roll year — the panel is 2007–2025, and the bars are flat, so nothing is missing *within* it.

2.2 Value: the assessor’s secured roll

Assessor Historical Secured Property Tax Rolls (wv5m-vpq2) is the value side: 3,934,467 parcel-years over 222,251 distinct parcels, carrying assessed land value, improvement value, fixtures, use code and definition, property class, year built, lot area, property area, number of units and stories — a parcel-year panel in the shape the hazard needs. Table 3 gives it.

The constraint is the period. **The roll runs 2007–2025 (19 years)**, and the corpus runs 1998–2026. The lower panel of Figure 1 shows the roll is flat inside its window — a median of 208,927 parcels a year, every year, with no hole — so this is a genuine start date and not the 2018 problem again. Of the 12,141 item parcel keys, 8,307 appear on the roll at some point, and 89.4% of items with a parcel key reach it. Of the item parcels that do appear, 6,168 appear in every one of the 19 years; the rest enter or leave, so the panel is unbalanced by construction and has to be treated that way.

What this costs. A parcel-year hazard can be estimated for 2007–2025. That is within the brief’s acceptance range (2005–2026 was asked for, 1998 was the ideal) but it excludes the first 9 years of the corpus, which is precisely the era in which the corpus memo’s substitution — denial falling to zero, conditioning tripling — begins. Anything that wants pre-period variation in assessed value will have to come from somewhere else, and §7 is the only candidate.

2.3 Zoning, and the one genuinely good surprise

The brief asked whether any *time-varying* zoning exists, expecting a current snapshot. It does, and the shape of the publication is the finding. Table 4 is the inventory.

- **9 vintages, spanning 1998–2008, are published *parcel-keyed***, one row per blklot with the zoning district, the height limit, the land use and, in the earlier years, unit counts and building square footage. These join to the item table on the same key as everything else in §2 and need no spatial work at all.
- **11 more — the remaining years and the three current layers (districts, height and bulk, special use) — are published as polygons, and are now joined.** An earlier draft of this section said the spatial join “should not be done accidentally.” It has now been done deliberately, and §3 reports it.
- **One year in the middle of the parcel-keyed run is not published in either form,** and Table 4 names it.

There is also a free current snapshot: `acdm-wktn` carries `zoning_code` and `zoning_district` on every parcel. So current zoning is parcel-keyed, 1998–2008 is parcel-keyed, and the years between them are the polygon-only gap in the middle. That is an odd shape but it is a far better one than a snapshot, and it means pre-period zoning variation is available for exactly the era the assessed values are not.

Height and bulk is worse. The historic height-and-bulk series is polygon-only in every year it is published; there is no parcel-keyed vintage (Table 13). Height therefore has to come either from the 1998–2008 zoning tables’ `height_lim` column or from a spatial join.

Land use (`c5ge-t6pj`) is available as a cross-check on the assessor’s use codes, with one wrinkle: its key is `mapblklot`, and 77 of its rows describe a *group* of parcels (`geography_type = multiple_parcels`, with a list of `mapblklot` values in one cell) rather than a single parcel. That is few enough to handle as an exception rather than as a design constraint.

3 The spatial join, and the panel it produces

The polygon layers are joined by point-in-polygon: a parcel centroid from `acdm-wktn` against each layer, in California Albers so areas are in square metres. Parcels enter and leave the panel on their own recorded dates (`date_map_add`, `date_map_drop`), so a hearing in 2003 is joined against the parcels that existed in 2003 rather than against today's.

Where the centroid rule is not enough. On the zoning layers the centroid places 98.7% of parcels; 2,563 fall in more than one polygon and 24,193 in none, and those go to a largest-area-overlap fallback using the parcel's own geometry. Table 15 reports both rates for every layer, because a centroid rule that silently mis-assigns boundary parcels is the kind of thing this memo exists to prevent.

A judgement call, stated. Special-use districts and neighbourhood impact-fee areas are *overlays*: a parcel legitimately sits in several of them or in none. An earlier pass treated their multi-matches as failures and sent more than two hundred thousand parcels to an area-overlap fallback that had no business running. Those families now carry a list of districts and no fallback, and a parcel in no fee area is recorded as being in no fee area rather than as a join failure.

The panel. Table 14 is the result. It holds 5,824,609 parcel-years over 1998–2026 and carries a zoning district on 95.1% of them. Of those rows, 4,098,871 come from the spatial join and 1,441,029 from the parcel-keyed tables. Two things in it are worth stating plainly:

- **The spatial route is more complete than the tabular one.** In the polygon years, 99.5% of the parcels alive that year get a zoning district. In the parcel-keyed years only 84.5% do, because those tables carry fewer parcels than the parcel layer says existed. The spatial join is not a second-best fallback for the years that happen to lack a table.
- **11 years carry a snapshot, and are flagged.** From 2016 onward no zoning vintage is published, so the current layer is carried forward with `is_snapshot` set. That is an assumption — Central SoMa and the 2018–2024 rezonings are inside that window — and it is recorded in the data rather than in a footnote, so no downstream code can use it without seeing it.

4 The risk set, defined

The previous section establishes that a parcel-year risk set *can* be built. It does not say what belongs in it, and that is not a detail: 236,552 parcels against a few thousand hearings is an extreme rare-event ratio, and a large share of those records are condominium interests that are not development sites.

Table 16 steps from every parcel-year down to a denominator, one restriction per row, each stated as a sentence, each reporting what it costs in events as well as in parcels.

- **Sub-parcel interests.** Condominiums, timeshares, co-ops and parking-stall interests are dropped, using the property class *in force that year* rather than the latest one — a parcel condominium-ised after its hearing must not be excluded retroactively. This is the expensive restriction: it discards 1,301 parcel-years in which the Commission actually heard something.
- **Public, institutional, church and school land.** A smaller cut.

- **No zoning in force.** Water, rights of way and the handful of parcel-years the panel cannot zone. Smaller still.

The decision. The preferred denominator is all three restrictions applied: **173,325 parcels, 4,261,894 parcel-years, 9,240 events, a rate of 21.7 per ten thousand** against 19.3 unrestricted. The rule is stated rather than tuned, and the brief’s instruction not to tune until the event loss is zero is followed: 1,301 hearings are lost to the condominium rule and stay lost.

Why that is acceptable, and where it is not. At the level of the item rather than the parcel-year, the denominator retains 94.5% of the 12,688 items that sit on a panel parcel-year (Table 17), and the loss is spread across request types rather than concentrated in one — no kind of case is being defined out of the sample. **But a paper about discretionary review volume should not use this denominator:** a DR on a condominium unit’s alteration permit is real Commission business and this rule removes the parcel it sits on. The denominator is built for the question of whether a *development site* produces a filing, and it should be used for that question only.

5 Filing dates and the conditional-use-to-permit bridge

5.1 Does the case number join?

The two Planning Department Records tables — y673-d69b (Non-Projects, 233,394 rows) and qvu5-m3a2 (Projects, 54,515 rows) — are the department’s own application record. Both identifiers were verified and both are unique on `record_id`. Table 5 tests the join three ways against all 8,958 distinct cases in the item table.

- **Raw string equality: 74.7%.**
- **Normalised (upper-case, whitespace stripped): 75.0%.** Normalisation buys almost nothing here, which is a small vindication of the extraction — the case numbers come out clean.
- **Suffix-stripped stem: 92.8%.** Stripping the letter suffix (2014.0400CUA → 2014.0400) is what does the work, because it is how a suffixed case reaches its umbrella project record. 8,838 of 8,958 cases join.

Correction, 2026-09-08: the records do go back that far. This section previously concluded that “the residual is a format, not a coverage rate, and no amount of fuzzy matching fixes it: the records simply do not go back that far under any key,” on the evidence that none of the two-digit-format cases matched. **The second half of that was wrong.** The format diagnosis was right; the conclusion drawn from it was not. Pulling every non-project record opened 1996–2002 shows 233,394-row table holding identifiers in the form 1998.505C — the department writes a four-digit year where the minutes print two. Expanding the year is a deterministic normalisation, not a fuzzy match, and it recovers **527 of the 530 such cases**, lifting the overall join to 98.7%. (The population is 530 rather than the 536 the earlier draft counted because expanding the year merges a handful of cases the minutes printed both ways — which is itself evidence that the expansion is doing what it claims.)

The one judgement call is the century, and it is resolved against the hearing year rather than a fixed pivot: a case heard in 1998 cannot have been filed in 2084. The rule is checked rather than

trusted — **report** counts the cases for which *both* centuries would have matched a real record, and Table 5’s caption carries that count. It is zero, which is why this is not labelled a fuzzy match anywhere it is used.

Figure 2 gives the join by hearing year. Of the modern-format cases only 92 fail; the era that looked unreachable is now the era with the second-best rate in the table.

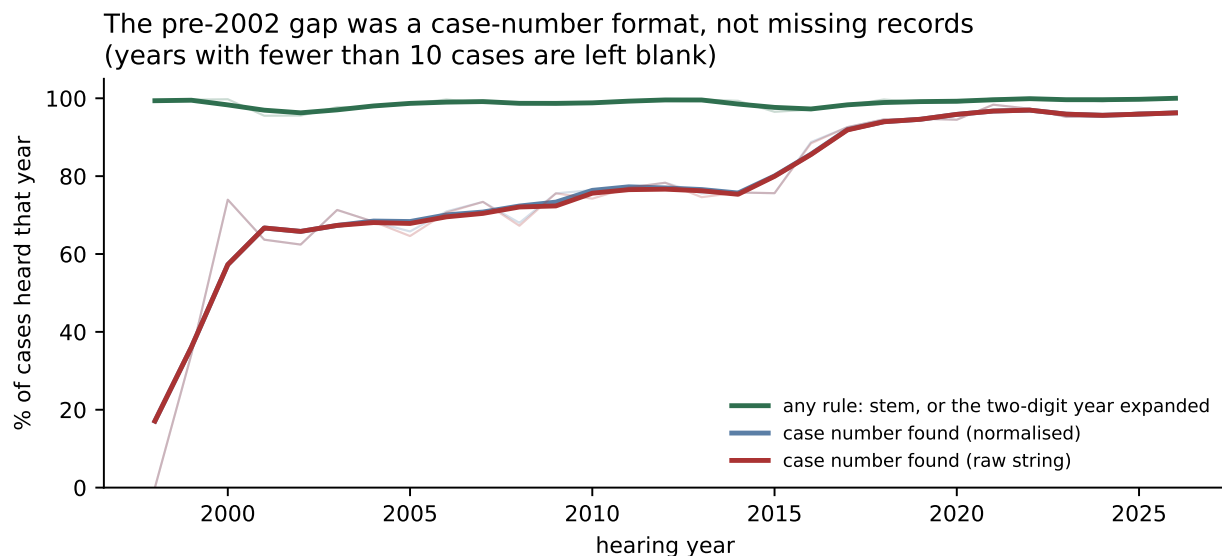


Figure 2: Share of cases heard each year whose case number is found in a Planning Department record. The red and blue lines collapse before 2002 because the minutes print a two-digit year there; the green line expands it against the hearing year and the collapse disappears. The records were always there.

5.2 What open_date buys: the wait before the hearing

open_date is populated on 100.0% of non-project records (Table 8), so every joined case has one. Collapsing both sides to one row per case — earliest open date, earliest hearing — gives the elapsed time a developer actually experiences before the Commission first looks at the project. Table 6 and Figure 3 report it.

- **Median 140 days.** 90th percentile 730; 99th 2314. Coverage 98.7% of all 8,958 cases.
- By request type the ordering is the one the theory wants: **variance 273 days, conditional use 168, discretionary review 101**. A discretionary review is filed against someone else’s pending permit and is heard quickly; a conditional use is the developer’s own application and waits.
- 0.4% of observations are negative — a record opened after the hearing. Those are follow-on records, and they are the reason the *earliest* open date of the case is the one used. They are reported rather than trimmed.

5.3 Two clocks, and which one the paper uses

A case’s stem can carry several records — a preliminary project assessment, an environmental filing, the entitlement application itself, follow-on actions. Taking the earliest of them is one clock; taking the earliest *entitlement* record is another, and they are not the same object. Table 20 reports both, and the gap between them as a distribution in its own right.

- **Earliest record of any type:** median 140 days, 90th percentile 730, on 98.7% of cases.
- **Earliest entitlement record:** median 135, 90th percentile 653, on 95.9%.
- **The gap:** median 0 days, 90th percentile 8, 99th 871. For nine cases in ten the earliest record *is* the entitlement record; the pre-application process is a long tail on a small minority, and it has not been measured here before.

The decision. The paper will use the entitlement clock. The real-options object is the decision to file the discretionary application whose outcome the Commission resolves; a preliminary project assessment is an inquiry that does not commit the developer to that hearing. The cost is 95.9% coverage against 98.7%, which is cheap, and the benefit is that the measure means one thing. The any-record clock is kept in the panel beside it, so the choice is testable rather than baked in.

Where the long tail comes from. It is not noise and it is not mis-dating. Median filing-to-hearing time rises monotonically with how many records share the stem: from 71 days for the 377 cases with a single record to 653 days for the 409 with more than ten. A stem that accumulates records is a project that has been through environmental review, revisions and re-applications, and its clock is long for a substantive reason. That answers the open item the previous draft left as a shrug.

This is the memo’s answer to the delay memo’s blind spot. That memo measures hearing-to-issuance at a 245-day median with a 768-day 90th percentile — both recomputed here from the artefact that memo wrote, and both reproduced exactly. The pre-hearing wait is a median of 140 days with a 90th percentile of 730: a shorter typical wait and an almost identical tail, and therefore a markedly larger ratio of tail to centre. A real-options model in which dispersion is what raises the filing threshold has been ignoring the more dispersed of the two waits, and starting its clock after the more dispersed one had already run.

5.4 The bridge, and a correction to the conditions memo

This was the highest-value question in the brief and the answer is yes.

`building_permits` is a field on both records tables, populated on 22.9% of non-project rows and 45.2% of project rows (Table 8). For each item, the permit numbers were read from the case’s own record, from its parent PRJ record where `parent_id` names one, and from any record sharing its stem; each was normalised to digits exactly as `analyze_permits.py` normalises a printed one — no zero-padding, because DBI does not pad the serial — and matched against the cached Building Permits table. Table 7 is the result.

- **Conditional use: 46.3% of 5,074 items reach a permit present in DBI**, against 0.3% that print a permit number in the minutes. The route the permit memo had was blind to this part of the docket almost completely; the bridge is not.

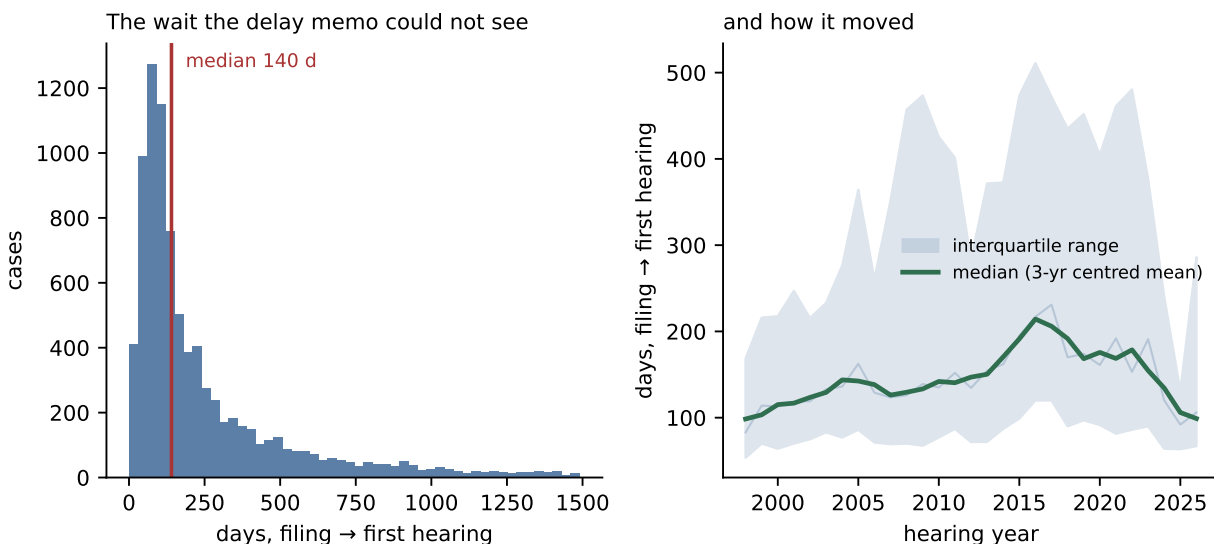


Figure 3: Left: the distribution of days from filing to first hearing, truncated at 1,500 days for display; the median line is computed on the untruncated data. Right: the median by hearing year with the interquartile range behind it, years with fewer than 10 cases omitted.

- Discretionary review reaches 71.8% through the bridge — *lower* than the 85.5% that print a number — so the two routes are complements, not substitutes. Taking either, 93.3% of discretionary reviews reach a permit.
- **Across the whole docket, 54.2% of items reach a building permit by one route or the other**, against 48.5% by the bridge alone and 22.1% by the printed number alone.
- Large project authorization (71.1%), variance (63.4%) and downtown project (58.3%) were all effectively unreachable before and are now reachable. Legislative items (planning code amendment, 7.6%) remain unreachable, correctly — they are not about a building.

Correction. The conditions memo, §5.1, concludes:

“The permit route reaches discretionary review (83.8% of DR items match a DBI permit) and essentially nothing else; the conditions route reaches conditional use and essentially nothing else. Between them they cover most of the docket, but they never cover the same item, so an analysis that wants both construction outcomes and condition content on one project cannot have them from these two sources.”

That conclusion no longer holds. 2,348 conditional-use items — the exact population the packet route reaches — now also reach a building permit in DBI. The non-overlap was a property of the route, not of the docket: the minutes do not print a permit number on a conditional use, but the Planning Department records one anyway. Construction outcomes and condition content *can* be had on the same project, for 46.3% of conditional uses. **The conditions memo has been amended accordingly:** §5.1 now carries a dated correction paragraph quoting the sentence it withdraws, and its §8 item 5 (“the exact complement of the permit route”) has been narrowed to a claim about what the minutes print rather than about what is knowable.

5.5 What else the records carry

Table 8 inventories the fields the item table does not have, with the share of rows on which each is actually populated. `close_date` (94.9%), `record_status` (100.0%), `applicant` (66.9%), `assigned_to_planner` (77.7%) and `applicant_org` (34.3%) are all substantially populated and all are new information. Two are not what their names suggest: `project_decision` is populated on 18.3% of project rows, so the department’s own decision field is not a substitute for the extraction, and `inclusionary` is a flag on 2.9%.

6 Project scale

Table 9 is the answer, and it contains the memo’s clearest instance of the difference between a field existing and a field carrying information.

The Projects table is the spine. Its `number_of_units_prop` reaches 99.0% of conditional-use items through the case-number stem, over a date range that starts long before any purpose-built pipeline dataset. But the field is never null and is *zero* on most rows: **it is non-zero on only 27.7% of conditional-use items**. Reporting the first number alone would be reporting the documentation. The right statement is that a unit count is *attached* to the great majority of conditional uses and is *informative* on well under a third of them — which is consistent with the docket, most of which is not residential construction, but which caps what a revenue measure built on it can do.

The purpose-built datasets are weaker than the brief hoped:

- **SF Development Pipeline** (6jgi-cpb4) does carry a planning case number — `case_no` — which is what the brief flagged as a potential P0 find. It is the PRJ umbrella number, so it reaches the item table only through the stem, and it is a **current snapshot**: 1,917 rows, the projects in the pipeline now. It reaches 4.5% of conditional-use items. It is a good source for scale on live projects and not a panel.
- **Housing Production 2005–present** (xdht-4php) carries `ppts_project_id` in the old YYYY.NNNN form, which joins to the item table’s stem, and `blocklot`. It reaches 19.1% of conditional-use items by parcel. It is completions, so it observes only projects that were built.
- **MOHCD Affordable Housing Pipeline** (aaxw-2cb8) carries `planning_case_number` explicitly and is 190 rows. It reaches 1.5% of conditional-use items. It is the right source for the affordability channel and it is small.
- **Dwelling Unit Completion Counts** (j67f-aayr) is keyed on the building permit number, so it is reachable only through §5.4’s bridge — which is now a reason it is reachable at all.

And none of it reaches the early docket well. Every purpose-built scale dataset begins in 2005 or later, and the consequence is measurable rather than rhetorical: a non-zero unit count reaches 20.3% of conditional uses heard before 2005 against 30.9% of those heard after.

7 Prices and rents

Table 10 is the fallback chain. The top of it is the finding.

CoreLogic / Cotality: the pull happened, and the extract is here. The brief asked whether the pull went through before Stanford access lapsed. It did, and the files are under the data root: `demand/` holds 76 files and 4.92 GB, including both the raw Cotality extracts and the cleaned parquet `demand_estimation/corelogic.py` built from them.

An earlier pass of this memo reported the whole `demand/` subtree as absent. It was wrong, and the reason is worth recording because it will recur. Dropbox’s macOS file provider keeps files *online-only*: the directory entry reports the full size while the file occupies zero blocks, and a directory that has never been materialised does not appear in a listing at all. `find` and `ls` both said nothing was there. `acquire_external_data.py sync` is now the check — it reports nominal bytes against bytes on disk and materialises what is dehydrated — and running it turned up 4.92 GB across 76 files, of which several gigabytes were placeholders. **The lesson is the one the repo already knows in another form: when the machine and the record disagree, check the source before believing the machine.**

What it is, and that it joins at the parcel. Table 10 and Figure 4 report it. The extract holds 447,883 San Francisco transactions, of which 415,854 are arms-length with a price and 320,323 are residential, over 174,102 distinct parcels.

The join is the `find`. Cotality prints San Francisco’s APN as `0836 003` — block and lot, in the same four-plus-three form DataSF uses — so stripping the separator yields `blklot` directly. **100.0% of San Francisco transactions carry a usable parcel key**, and the price therefore joins to the item table on exactly the key the risk set of §2 is built on. This is a parcel-level price, not a market-level index, and it is the difference between having $\Pi(s)$ and having a proxy for it.

What that buys, measured rather than asserted:

- **5,620 of the 8,980 distinct parcels the item table touches (62.6%) have at least one arms-length sale.**
- **63.0% of items with a parcel key** sit on a parcel that has traded at some point — 8,574 items.
- **25.5% sit on a parcel that traded within 3 years of the hearing** (3,477 items). That is the number that matters for a filing threshold: it is the share of decisions for which a *contemporaneous* market price on the actual parcel is observable, rather than a neighbourhood average.

The right panel of Figure 4 shows the first of those rising from roughly half to three quarters over the period, as more of the housing stock turns over at least once inside the window the extract covers, while the contemporaneous-trade share sits flat near a quarter. The flatness is the honest read: turnover, not data coverage, is what caps it.

7.1 A price on every parcel-year

25.5% is a fine coverage rate for validating a price and an unusable one for a hazard over the whole risk set. The 408,119 transactions plus the assessor’s characteristics are enough to *predict* where they cannot observe.

The surface is a neighbourhood-by-year effect — 2,162 cells, using the analysis neighbourhood the parcel layer already carries — absorbed out of log price, plus a hedonic on lot area, building area, units, stories, year built and use code, fitted on the within-cell variation. A repeat-sales specification is feasible on the 62.4% of traded parcels that trade more than once, and was compared against

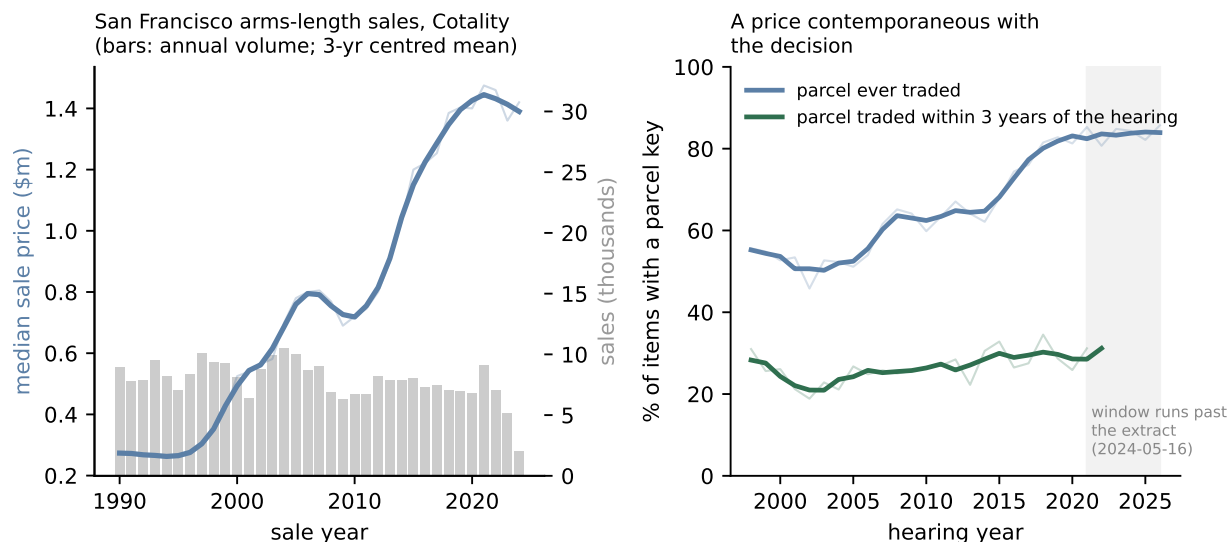


Figure 4: Left: median arms-length sale price in San Francisco with annual volume behind it, from the year the extract becomes dense. Right: the share of items carrying a parcel key whose parcel has ever traded, and whose parcel traded within 3 years of the hearing. The shaded band is the hearing years whose forward window runs past the end of the extract; the contemporaneous series is not drawn there, because a fall would be an artefact of where the file stops.

it; the hedonic is what is used, because a repeat-sales index is undefined on every parcel that has never traded, and those are exactly the parcels the prediction exists for.

Table 19 reports the fit, scored out of sample on a held-out fifth (82,194 transactions):

- **RMSE 0.69 in log price**, R^2 0.72. That is a wide interval and it is stated as one: this is a price *surface*, not an appraisal.
- **RMSE 0.61 on parcels that trade at most once** — *better* than the pooled figure, which is the direction that matters, because those are the parcels being imputed for.
- Bias -0.037 in logs on the parcel-years where an observed price also exists, and a correlation of 0.73 between predicted and observed.

One error worth recording. The first version of this surface carried a +0.90 log bias — a factor of two and a half on every price. The model is fitted on deviations from the cell mean, and the prediction was adding the raw characteristic *level* instead of its deviation from the cell’s mean characteristics. It reproduced sensible-looking time series and was wrong by a factor of two and a half, which is the argument for scoring a prediction against held-out data rather than looking at its plot.

What is cached. 3,948,435 parcel-years in the denominator carry a price; 2.7% of them are observed sales and the rest are imputed, in separate columns with an `is_observed` flag, so no downstream code can confuse the two. Assessed value enters as a covariate and never as the price, because Proposition 13 makes it a function of tenure.

Where it stops. The extract ends 2024-05-16, two years short of the corpus, and the series is only dense from 1968 (a handful of deeds carry placeholder dates back to 1900). The last three hearing years therefore cannot have a 3-year forward window and are drawn as a gap in Figure 4 rather than as a decline — the same rule the corpus memo applies to 2018.

CoStar is an access question and is reported as one. Nothing was attempted and nothing was scraped.

What is cached instead. Zillow’s research files, filtered to San Francisco County and stored with the national row counts recorded so the filter is auditable: **ZHVI on 26 ZIP codes, 2000-01 to 2026-07**, and **ZORI on 24 ZIP codes, 2015-01 to 2026-07**. These are market-level indices; the item table has no ZIP, so using them means going parcel → address → ZIP, or falling back to the supervisor district and analysis neighbourhood the parcel layer already carries. They are a price *environment*, not a project price.

The fallback. Assessed land and improvement value from §2, parcel-year, 2007–2025, universal within that window. It is a weak price: Proposition 13 makes assessed value a function of how long the current owner has held the parcel rather than of what the parcel is worth. It is worth having with that caveat stated in every table it appears in, and it is not a substitute for a transaction price.

8 Fee and exaction schedules

The conditions memo finds that most staff-imposed conditions are obligation-imposing rather than project-changing, and that the largest recurring headings — Transportation Sustainability Fee, inclusionary, impact fees — have posted dollar schedules. This section converts those headings into dollars.

The document. The **San Francisco Citywide Development Impact Fee Register** is a multi-page landscape schedule, republished each January, giving for every fee: the city area subject to it, the ordinance reference, the uses it applies to, the rate, the threshold, the controlling entity, the unit, and the effective date. It is exactly the object the brief asked for.

The problem, and the route around it. The register is republished at *one URL*, so the department’s site only ever holds the current year. 8 dated copies were assembled — 2019 through 2026 — from the Wayback Machine’s snapshots of that URL and from the year-stamped URLs the department left behind. **No snapshot exists before 2019-09-15**, under that URL or any other tried, so the fee panel starts at 2019 (Table 13).

The parse. Table 11 reports what came out of each register: 1,341 dated rate observations in total, keyed to the Planning Code (or State Education Code) section of the register row they sit in, and to the land-use or size band the register prints on the same line. The section is the key because it is the only column that survives PDF text extraction intact; the fee *name* wraps across several lines and the columns interleave. Effective dates are read from the register’s own cover line, not inferred from when the file was retrieved.

Table 12 is the result: **a rate keyed by year, by fee and, where the register labels it, by land use**. The Transportation Sustainability Fee for residential projects over 99 units runs \$10.29

per gross square foot in 2019 to \$13.39 in 2026; the inclusionary fee under §415 runs \$199.50 to \$249.66 per square foot of residential gross floor area. A named condition heading in a Commission motion can now be priced.

8.1 Does the schedule identify anything?

A citywide schedule that drifts up a few per cent each January is absorbed by a year fixed effect, and if that were all the register held it would be a control rather than a source of variation. It is not all it holds. Parsing the land-use and size bands into a structured schedule — 455 rows keyed on year, section, land use and band — and decomposing the variation in the log rate gives Table 18:

- **Year alone absorbs 3.0%.** Almost nothing: the dispersion in this schedule is across fees, not over time. The inclusionary fee is two orders of magnitude larger than the child-care fee, and no year effect touches that.
- **Year and Planning Code section together absorb 83.6%, leaving 16.4% within a year and a fee** — across land uses and size bands.
- Conditioning on the land-use or band label too leaves 9.4%, which is the tier and district variation.

So fees are a usable source of cross-sectional variation and not a time-series one, and the design that uses them is a discontinuity one: the Transportation Sustainability Fee steps at 99 residential units and §415 at ten, and those are steps in the cost of filing that a real-options threshold has a direct prediction about. The size bands parse on 2.6% of rate observations — low, but concentrated on exactly the fees with a threshold, which are the ones the design needs.

What is not here. The register’s *district* dimension is a free-text column (“Eastern Neighborhoods Mixed Use Districts: RED, RED-MX, SPD, MUG...”) and is not parsed into a key beyond a token match. Of 42 register rows, 11 name a zoning district the panel recognises (26.2%); the rest are citywide or name an area plan rather than a zoning district, so a low rate here is mostly correct rather than a parse failure — but it is a rate, not a claim, and rows that do not parse are counted rather than dropped. Neighborhood-Specific Impact Fee Areas (`ntc3-dd64`, 33 rows) *is* now joined to parcels through §3, so the geographic side of the fee is on the parcel-year panel. The inclusionary *requirement percentages* under §415 — as distinct from the in-lieu fee — and their amendment history are legislative and are not in the register; that remains open.

9 The hand pass

320 items carry a parcel key that does not join. 34 have the block and lot transposed — a real extraction error, counted and *not* silently corrected, because swapping them would be guessing. 21 name no block the city has ever recorded. The remaining 265 name a block that exists and a lot that does not, and 50 of those were drawn at random and read against the minutes text the extraction saw. Table 22 reports the classification and `handpass_sample.csv`, beside this memo, carries the sample so it can be checked.

The answer is clean and it is not what was expected. 46 of 50 (92%) are cases where **the extraction matches the minutes exactly** and the lot is simply absent from the parcel layer — typically a lot that was merged or resubdivided before the layer’s mid-1998 epoch, so the recorder’s retired set never carried it. There are 0 extraction errors in the sample. The previous draft left this open on the suspicion that the minutes had been mis-read; they had not.

10 Established, and not established

Established.

1. The parcel key joins on 97.7% of items carrying one, once the lot's digits rather than the whole token are padded and multi-block items are expanded. *This narrows the previous claim, which was about a risk set: what is established here is the join.*
2. A parcel-year zoning panel exists over 1998–2026: 5,824,609 rows, zoned on 95.1%, of which 4,098,871 rows come from a spatial join with a stated centroid rule and a stated fallback. 11 years carry a carried-forward snapshot and are flagged as such in the data.
3. A denominator is *defined*: 173,325 parcels, 4,261,894 parcel-years, 9,240 events, 21.7 per ten thousand, under three stated restrictions, one of which costs 1,301 real hearings.
4. The value side of the risk set is 2007–2025 and no earlier.
5. The case number joins on 98.7% of cases once the two-digit year is expanded against the hearing year, with zero century ambiguity. *This resolves the previous version of this item, which put the pre-2002 era beyond reach.*
6. Filing to first hearing is measurable on two clocks that differ by a median of 0 days; the paper uses the entitlement clock, and the long tail is explained by the number of records under the stem rather than by mis-dating.
7. The **building permits** bridge reaches 46.3% of conditional-use items and 54.2% of the docket by some route. The conditions memo's non-overlap claim (§5.1) is overturned.
8. Prices exist on every parcel-year in the denominator, observed on 2.7% and imputed on the rest at an out-of-sample RMSE of 0.69 in logs, flagged apart.
9. Fees carry 16.4% of their log-rate variation within a year and a section, so they identify cross-sectionally; a year effect absorbs 3.0%.
10. 92% of the block-matched, lot-unmatched residual is the parcel layer's coverage, not an extraction error.

Not established.

1. Whether the 34 transposed items should be repaired. The swap joins, but a rule that swaps block and lot whenever the swap joins will also swap correct pairs by coincidence; the count is reported and the repair is not made.
2. Whether the condominium restriction is right. It is defensible for a filing model and wrong for a discretionary-review model, and 1,301 events sit on the boundary.
3. Whether zoning after 2016 is what the panel says it is. It is the current layer carried backward, which the flag records and no analysis should ignore.
4. Whether the price surface is unbiased *conditional on filing*. It is unbiased on the parcel-years where a sale is observed, but a parcel that trades near its hearing may trade because of it, and nothing here corrects for selection on unobservables at the parcel level. The imputation fixes coverage, not endogeneity.

5. Whether the permit a conditional use reaches through the bridge is the permit the hearing was about.
6. Anything about fee levels before 2019.
7. Anything causal.

11 Limitations

The coverage is not missing at random. Table 21 crosses every headline rate against request type, outcome, conditioning and era. Four cells need naming rather than leaving for the reader:

- **Conditioned items carry a usable parcel key on 92.9% against 78.5% for unconditioned ones.** That is a fourteen-point differential in a variable that is close to the outcome of interest, and it means the parcel-keyed analysis is run on a sample tilted towards conditioned decisions. It is the most consequential number in that table.
- **Legislative items barely have parcels at all:** planning code amendments join on 9.3%. That is correct — they are not about a site — but it means any parcel-level denominator silently excludes the legislative docket, and the exclusion should be explicit rather than emergent.
- **The outcome category ‘other’ joins at 50.8%** against 82.6% for approved and more for the rest. Items whose disposition the extraction could not classify are also items whose parcel it could not read; the two failures share a cause, which is a thin or malformed block.
- **Scale and permit coverage rise steeply with era:** non-zero unit counts run 23.8% in the earliest era against 47.6% in the latest, and the permit bridge 41.9% against 61.5%. Any specification using those variables is weighted towards the recent docket whether or not it says so.

The parcel join itself is roughly flat across eras (85.2%, 78.0%, 81.4%), so this is not one era effect wearing several hats.

Coverage rates resting on a small denominator. The MOHCD pipeline is 190 rows and reaches 1.5% of conditional-use items — that is a handful of items, and a rate computed on it should not be reported to one decimal place anywhere but the table it comes from. The Development Pipeline’s 4.5% rests on 1,917 rows. The request-type breakdown in Table 7 suppresses types with fewer than 100 items, and Table 6 suppresses groups with fewer than 10 cases with a lag.

Current snapshots that are not panels. The Development Pipeline is the projects in the pipeline *now*; a project entitled a decade ago and long since built is not in it. The current zoning, height-and-bulk and special-use layers are snapshots. `acdm-wktn`’s `zoning_code` is current zoning attached to every parcel including retired ones, which is a snapshot wearing a panel’s clothes and must not be used as the zoning *at* the time of an early hearing — that is what the 1998–2008 vintages are for. San Francisco Land Use is one `data_as_of` date.

Periods that do not line up — restated, because the spatial join moved them. The earlier version of this paragraph said that a specification wanting both an assessed value and the zoning in force was confined to a two-year overlap “unless the polygon zoning years are joined spatially.” They have been. **Zoning now spans 1998–2026** (1999 excepted), so the binding constraint is no longer zoning. What remains:

- Assessed values: 2007–2025. **This is now the binding constraint on any specification needing an assessed value**, and it costs the first 9 years of the corpus.
- Transaction prices: 1968–2024, stopping at 2024-05-16, two years short of the corpus. The imputed price panel inherits that end date.
- Fee rates: 2019–2026.
- Zoning after 2016 is a carried-forward snapshot, so it spans the corpus without *observing* it in those years.

The usable full-information window is therefore 2007–2024 — the intersection of the assessed values and the transaction prices — with fees available only over its second half. That is a longer window than the previous draft implied and a shorter one than the corpus.

Access questions left open. CoStar (BU Questrom holdings, unchecked by design). The §415 inclusionary requirement percentages and their amendment history. Fee registers before 2019. CoreLogic is *not* on this list any more — the extract is on disk — but the licence is Stanford’s and lapsed, so the file cannot be refreshed past 2024-05-16 without new access. Treat what is cached as frozen.

Source quirks worth carrying. Dwelling Unit Completion Counts contains a `date_issued` in 2205, marked (*sic*) in Tables 1 and 9; it is a typo in the source, not the end of the file’s coverage. The MOHCD pipeline’s dates run to 2030 because they are estimated construction completions, which is correct. 14 of the 19 cached sources are not unique on their intended key and are marked so in Table 1. The Planning Department’s non-project table has an `open_date` in 1920, which is real in the sense that the row says so and has not been audited.

Two model choices that are assumptions, not measurements. The largest-overlap fallback assigns a boundary parcel to whichever district covers most of it, which is a rule and not a fact; and the price surface’s neighbourhood-by-year cells assume that within a neighbourhood and a year, characteristics price linearly. Both are stated here because neither is testable against anything this memo holds.

12 Reproducing this

`code/commission_minutes_processing/acquire_external_data.py` runs in three stages, each cached so no network stage repeats by accident:

```
python acquire_external_data.py sync           # materialise online-only files
python acquire_external_data.py probe          # resolve every identifier, live
python acquire_external_data.py fetch          # download and cache each source
python acquire_external_data.py fetch --refresh # re-download everything
```

```
python acquire_external_data.py fetch --only polygons    # the GeoJSON layers
python acquire_external_data.py spatial                 # point-in-polygon -> the panel
python acquire_external_data.py spatial --refresh
python acquire_external_data.py report                  # joins, figures, tables, README
```

`sync` materialises every online-only file under `$MFHR_DATA_ROOT/demand/` and prints nominal bytes against bytes on disk. Run it first: it is the difference between a subtree that is missing and one that Dropbox has simply not downloaded, and this memo got that wrong once.

`probe` writes `$MFHR_DATA_ROOT/external/_acquisition/probe.json` — the resolved identifier, row count, column list and update time for every dataset, the catalogue searches that produced them, and an HTTP check on every non-Socrata file. `fetch` writes one directory per source under `$MFHR_DATA_ROOT/external/` (parcels, assessor, zoning, planning_records, pipeline, prices, fees) and skips anything already on disk. `fetch --only corelogic` is the exception that downloads nothing: it slices the licensed Cotality extract already under `demand/` down to San Francisco, attaches the parcel key, and caches the result, so `report` does not re-read two gigabytes. `spatial` builds the parcel-year zoning panel (`$MFHR_DATA_ROOT/external/zoning/parcel_zoning_panel.parquet`) and a per-layer coverage file beside it; it is the one stage that needs `geopandas`, and it takes a few minutes. `report` additionally writes the imputed price panel and `handpass_sample.csv` next to this memo.

`report` regenerates the four figures, `tables/acquisition_tables.tex`, `tables/acquisition_macros.tex`, and `external/README.md`, which records per source the resolved identifier, retrieval date and row count so a later run can tell whether the cache is stale.

The item table is the same extraction the other memos read (`$MFHR_DATA_ROOT/extraction/corpus_v2.g3/clean/*.jsonl`), and the permit normalisation and the printed-permit parser are imported from `analyze_permits.py` rather than re-implemented, so the two memos cannot report different numbers for the same question.

No number in this memo is hand-placed, and here that is literal rather than conventional. Every measurement in the prose is a `\newcommand` defined in `tables/acquisition_macros.tex`, written by `report`; the tables are `tables/acquisition_tables.tex`, written by the same function; both are `\input` at the top of the `.tex`. Changing a figure in this document means changing the script.

The only numerals typed into the `.tex` are: section and item cross-references; identifiers and dataset names (2014.0400CUA, 98.426D, Housing Production 2005–present); the register’s own printed land-use bands (“over 99 units”); named prior facts (Proposition 13, the 2018 hole in the meeting corpus); the brief’s own $\geq 95\%$ acceptance criterion; and three figures quoted from sibling memos, each attributed where it appears — the corpus memo’s 8,295 distinct parcels, the permit memo’s 145,795 duplicate `permit_number` rows, and the 83.8% inside the block quotation from the conditions memo. To those this revision adds four figures quoted from the passages it withdraws — the 89.5% parcel join, the 1,232 block-matched items, the 0 of 536 pre-2002 matches, and the +0.90 log bias of the first price surface — plus the 27,822 lettered parcels that explain the first two. A withdrawn number has to be quotable for the correction to mean anything. The delay memo’s hearing-to-issuance median is *not* in that list: `report` recomputes it from `permit_matches.csv` and it appears here as `\acqDelayIssueMedian`, which is one way of checking that this memo and that one still agree.